



Amazon ML Summer School 2025 Assessment

03 Aug 2026 | 10:30 AM Morning Shift

Section 1: MCQ Section

Note to Student: Please solve the following questions. Choose the most appropriate option for each multiple-choice question.

Question 1: Simplify $(x + y)^2 - (x - y)^2$

- A) $4xy$
- B) $4x^2$
- C) $4y^2$
- D) $2x^2 + 2y^2$

Question 2: If $\frac{3x + 2}{(x - 2)(x + 3)} = \frac{A}{x - 2} + \frac{B}{x + 3}$, find the value of A.

- A) $-5/9$
- B) $8/5$
- C) 1
- D) 0

Question 3: For $f(x, y) = x^2 + y^2 - 4x - 6y$, what are the coordinates of the minimum point?

- A) (2, 3)
- B) (-2, -3)
- C) (4, 6)
- D) (0, 0)

Question 4: In a group of 100 students, 60 play cricket, 40 play football, and 20 play both. How many students play neither cricket nor football?

- A) 10
- B) 20
- C) 30
- D) 40

Question 5: You are working on a dataset where the variable `user_location` contains over 10,000 unique values. Which encoding strategy is most appropriate?

- A) Label Encoding
- B) One-Hot Encoding
- C) StandardScaler
- D) Target/Mean Encoding



Question 6: You are training a time-series model on stock prices. Which of the following cross-validation techniques is most appropriate?

- A) K-Fold Cross Validation
- B) Stratified K-Fold
- C) Leave-One-Out
- D) TimeSeriesSplit

Question 7: Two dices are rolled randomly on a sheet. What is the probability that two particular faces land up next to each other?

- A) $\frac{1}{6}$
- B) $\frac{1}{3}$
- C) $\frac{1}{2}$
- D) $\frac{1}{18}$

Question 8: A committee of 3 people is selected at random from a group of 5 men and 4 women. What is the probability that exactly 2 women are selected?

- A) $\frac{10}{84}$
- B) $\frac{40}{84}$
- C) $\frac{30}{84}$
- D) $\frac{20}{84}$

Question 9: The Posterior distribution is proportional to:

- A) Prior / Likelihood
- B) Likelihood / Prior
- C) Likelihood $\times$ Prior
- D) Prior + Likelihood

Question 10: Which of the following best describes the concept of Bayesian Inference in ML?

- A) Estimating parameters using maximum likelihood
- B) Minimizing the hinge loss function
- C) Updating prior beliefs with observed data using Bayes' Theorem
- D) Applying gradient descent to probabilistic distributions

Question 11: Which of the following is a valid assumption of PCA (Principal Component Analysis)?

- A) The principal components maximize the explained variance
- B) Variance captures the important structure
- C) The components are linearly uncorrelated
- D) The data has a Gaussian Distribution

Question 12: In Reinforcement Learning, what does a policy define?

- A) The state of the environment
- B) The reward structure
- C) The probability distribution over actions given a state
- D) The discount factor



Question 13: How does Gradient Boosting differ from Bagging?

- A) Boosting cannot be used with decision trees
- B) Boosting builds models independently, Bagging builds sequentially
- C) Bagging always produces better accuracy
- D) Boosting reduces bias, Bagging reduces variance

Question 14: Which of the following is a problem commonly addressed by using Batch Normalization in neural networks?

- A) Internal covariate shift
- B) Gradient vanishing
- C) Model underfitting
- D) Overfitting due to large weights

Question 15: Which cost function is most appropriate for a binary classification task?

- A) Binary Cross-Entropy
 - B) Hinge Loss
 - C) Categorical Cross-Entropy
 - D) Mean Squared Error (MSE)
-

Section 2: Coding Section

TEST INSTRUCTIONS

- **GENERAL**

- This is a timed test. Please keep track of the time remaining at the top of the page.
- If you feel the question is unclear, try out different inputs and check the expected output for each of them.

- **CODE IMPLEMENTATION**

- You only need to implement the given function.
- Do not read input; instead use the arguments to the function.
- Do not print the output; instead return values as specified.

- **PLAGIARISM**

- Your code submission will be checked for plagiarism after you're finished.
 - To prevent plagiarism, we discourage switching tabs or windows during the test.
-



Q1. Alien Invasion (Space Cannon)

Problem Description

The alien ships are approaching Earth at a constant speed. The initial distances of the ships are provided in an array A , along with the constant speed of each ship in an array B . You have a space cannon that, once fully charged, can destroy a single alien ship. However, the cannon takes exactly 1 minute to charge.

You lose if any alien ship reaches Earth. If a ship reaches Earth at the exact moment the cannon is fully charged, it counts as a destroy. Return the maximum number of alien ships that you can destroy before you lose, or N if you can destroy all the ships before they reach Earth.

Problem Constraints

$$1 \leq N \leq 10^5$$
$$A[i], B[i] \leq 10^5$$

Input Format

First Argument is an Integer Array A , denoting the initial distances.
Second Argument is an Integer Array B , denoting the speed.

Output Format

Return an Integer, denoting the maximum number of alien ships that you can destroy before you lose.

Solution Template:

```
1 int Solution::spacecannon(vector<int> &A, vector<int> &B) {  
2     // Write your logic here  
3  
4 }
```

Q2. Stacking Money

Problem Description

After a long fight, Wanda and Vision settle down to a normal life. One day, Vision was stacking coins. He made different stacks of coins represented in the form of an integer array A , where each stack had $A[i]$ coins.

Vision asked Wanda the following question: Find the sum of the difference of maximum and minimum number of coins over all subarrays of A .

Problem Constraints

$$1 \leq |A| \leq 10^5$$
$$1 \leq A[i] \leq 10^9$$

Input Format

The first argument is the integer array A .

Output Format

Return a single long integer denoting the value Vision asked for.

Example Input

Input 1: $A = [1, 2, 3]$



Example Output

Output 1: 4

Example Explanation

Explanation 1:

The subarrays along with their (max - min) differences are:

[1] $\rightarrow 1 - 1 = 0$

[2] $\rightarrow 2 - 2 = 0$

[3] $\rightarrow 3 - 3 = 0$

[1, 2] $\rightarrow 2 - 1 = 1$

[2, 3] $\rightarrow 3 - 2 = 1$

[1, 2, 3] $\rightarrow 3 - 1 = 2$

The sum over all subarrays is $0 + 0 + 0 + 1 + 1 + 2 = 4$.

Solution Template:

```
1 long Solution::stackingMoney(vector<int> &A) {  
2     // Write your logic here  
3  
4 }
```

i Important Note

These are the **real assessment questions from the Amazon ML Summer School 2025 (Morning Shift 10:30 AM)**. Please note that Amazon does not officially release these questions. The questions have been collected from candidates who appeared for the assessment, so there may be slight variations in numerical values or wording. Considerable effort has gone into compiling and organizing these questions, and they are intended purely for learning and assessment practice purposes.

For more content, solved papers, and guidance, visit my GitHub: <https://github.com/cu-sanjay>
If you find this document helpful, please star the repository and follow the account.

All rights are reserved by **Amazon** and the respective examination authorities.
Thank you. I wish you the best for the ML Summer School exam and program.

